NISHAN PAUL

Software Engineer

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EXPERIENCE

Vertical Innovations Ltd. | 2024-2026 | Fullstack Developer

Architected a Next.js/FastAPI smart dispensing system with RFID authentication and live inventory telemetry. Engineered real-time WebSocket sync and optimized PostgreSQL to track high-volume usage logs for critical hygiene compliance.

Dreams For Tomorrow | 2023-2023 | Contract Developer

Built a multi-tenant Laravel CMS and blogging platform, enabling secure resource sharing and collaborative forums for educational environments.

Tech Cloud Ltd. | 2022-2022 | Internship

Architected a Laravel-based accounting ecosystem, automating real-time financial tracking and reporting to maximize operational accuracy.

PROJECTS

Barcody - Barcode Scanner for Anything

Next.js | NestJS | PostgreSQL | Redis | TypeORM | Socket.io | BullMQ

Universal Barcode Intelligence & Scanning Suite. High-performance monorepo for scanning and analyzing barcodes at scale. Features real-time sync, Tailscale private networking, and Docker containers. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under barcody.kaiofficial.xyz

Markify - Markdown to PDF Converter

Next.js | Zustand | Framer Motion | Shadcn UI | PostgreSQL | Docker

A simple and fast Markdown-to-PDF converter that transforms .md files into clean, professional PDF documents. Supports code formatting, structured content, and customizable output for developers. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under markify.kaiofficial.xyz

Flick - P2P File Sharing

Next.js | React | TypeScript | Tailwind CSS | WebRTC | PeerJS

Instant peer-to-peer file sharing app built with Next.js and WebRTC. No server storage, no accounts — just share a room code and transfer files directly between devices. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under flick.kaiofficial.xyz

EDUCATION

MSc in Cybersecurity | 2025 –
Dhaka University
Dhaka, Bangladesh

BSc in CSE | 2017 – 2024
Chittagong University of Engineering and Technology (CUET)
Chittagong, Bangladesh

HSC in Science | 2014 - 2016
Notre Dame College
Dhaka, Bangladesh

SSC in Science | 2004 - 2014
Motijheel Govt. Boys' High School
Dhaka, Bangladesh

LANGUAGES

JavaScript (ES6+) | TypeScript | Python | Java | C# | C/C++

FRONTEND

Next.js | React | Tailwind CSS | Shadcn UI | Zustand | Framer Motion

BACKEND

NestJS | Node.js | Redis | TypeORM | Socket.io | BullMQ | Swagger

DATABASE

PostgreSQL | SQL Server | SQLite

DEVOPS

Docker | Docker Compose | Containerization | CI/CD (GitHub Actions) | Linux Systems | Nginx / Reverse Proxy | VPS Deployment | SSL/TLS (Certbot)

PROJECTS

CMatrix - Agentic Red Team

FastAPI | Next.js | LangGraph | LangChain | PostgreSQL | Qdrant | Celery

Agentic red team for running controlled, human-in-the-loop security simulations across modern applications and infrastructure. It combines multi-agent orchestration, approval gates, monitoring, and audit-friendly workflows to help teams test defenses safely, repeatably, and at scale. Hosted on a Linux VPS using Docker Compose, Nginx reverse proxy, and Certbot SSL under cmatrix.kaiofficial.xyz

OpenCV Visualizer

Python | OpenCV | NumPy | Computer Vision | Image Processing | Tkinter

A comprehensive toolkit for Computer Vision mastery using OpenCV and Python. This repository features 28+ modular implementations of image processing techniques, feature detection algorithms, and real-time video analysis tools (including Lane Detection & Invisible Cloak), all accessible through a custom-built, interactive Tkinter GUI.

PROBLEM SOLVING

Dynamic Programming (10) | Greedy (38) | Strings (16) | Hashing (2) | Bit Manipulation (2) | Graphs (3) | Algorithms (27) | Data Structure (4) | Math (54) | Implementation (89)

- Codeforces | 236 solved
- LightOJ | 89 solved
- Virtual Judge | 81 solved
- UVA | 14 solved

ACHIEVEMENT

Vivasoft AI Hackathon 2025

Team: KAI **Ranking:** Top 10

CUET Career Club – Competitive Programming 2019

Team: CUET_IRON_BAT **Ranking:** 02

Techtronics 2018 IUPC – Honorable Mention

Team: CUET_IRON_BAT **Ranking:** Top 50

ICPC Dhaka Regional 2017 – Online Preliminary Contest

Team: iRON-BAT **Ranking:** 308

CUET CSE FEST 2017 – IUPC

Team: CUET_iron_bat **Ranking:** 20

AI ENGINEERING

LangGraph | LangChain | FastAPI | Qdrant | Celery

DEV TOOLS

Git / GitHub | Pre-commit Hooks (Husky) | Build Tools (Vite / Turborepo) | Secure Auth (OAuth 2.0 / JWT)

AI TOOLS

Antigravity | Cursor | Kilo Code | Supermaven | Gemini CLI | Qwen CLI

TESTING

Automated Testing (Jest) | Postman | ESLint | Prettier | Mesh Networking (Tailscale)

CERTIFICATES

- Getting Started with Python
- Python Data Structures
- Managing Machine Learning Projects with Google Cloud
- AI For Everyone
- Foundations in Digital Forensic with Magnet Axiom
- Workshop on IoT by Planeter Ltd.

BLOGS

- Backtracking Technique & N Queen Problem
- Tiling a Defective Chessboard Using Divide & Conquer Technique
- Setup Flutter On VScode